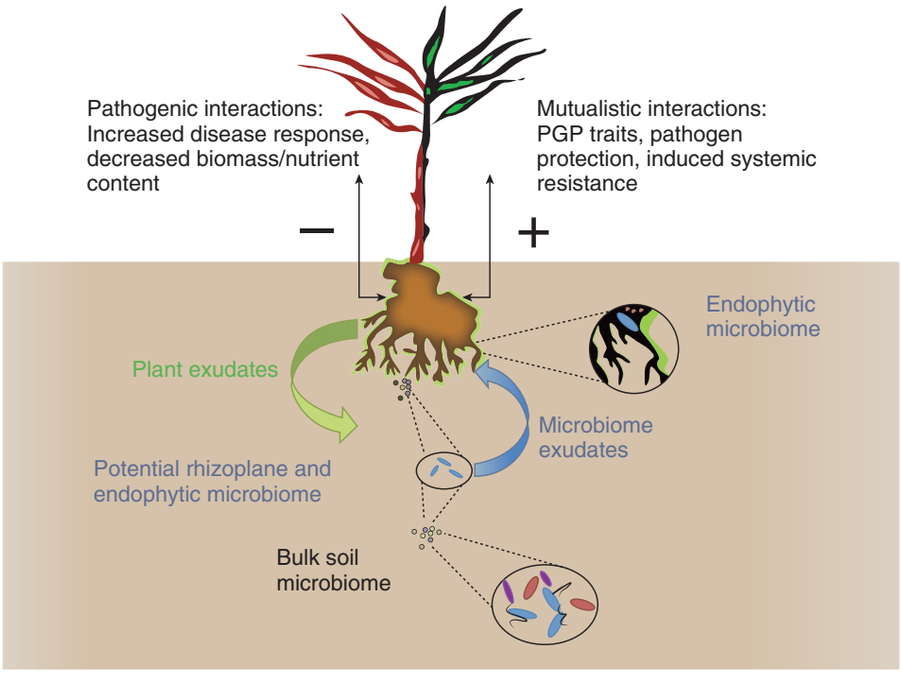


Figure 10.6: Effect of Rhizosphere in plant growth

Infobits

Bioleaching:

Soil microorganisms are very closely involved as catalytic agents in many geological processes. These include mineral formation, mineral degradation, sedimentation and geochemical cycling. In recent years, a new discipline of mineral science namely bio-hydrometallurgy or microbial mining (mining with microbes) is rapidly growing. Broadly speaking, bio-hydrometallurgy deals with the application of biotechnology in mining industry. In fact, microorganisms can be successfully used for the extraction of metals (Example: copper, zinc, cobalt, lead, uranium) from low grade ores. Mining with microbes is both economical and environmental friendly.

10 to 20 or sometimes more) than with actinomycetes or fungi (Figure 10.6). From the agronomic point of view, the abundance of Nitrogen fixing and Phosphate solublising bacteria in the rhizosphere of crop plants assumes a natural significance.

It has been reported that amino acid requiring bacteria exist in the rhizosphere in large numbers than in the root free soil. The rhizosphere effect improves the physiological conditions of the plant and ultimately result in higher yield.

10.7 Phyllosphere

The term "Phyllosphere" was coined by the Dutch Microbiologist Ruinen. The leaf surface has been termed as Phylloplane and the zone on leaves inhabited by the microorganisms as Phyllosphere (Figure 10.7). In forest vegetation, thick microbial epiphytic

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